Scalability & Future Plan

Date	02 October 2026
Team ID	SWTID-2026-3981
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	**Local In-Memory Execution & Inferences** Running the LaMini-Flan-T5-783M model locally relies heavily on host CPU/RAM resources.	Slows down request processing under multiple concurrent user requests.	High
2	**Stateless Backend & Lack of Persistence** No database integrated into the FastAPI backend (main.py).	Quiz results, QA history, and learning path progress cannot be saved across sessions.	Medium
3	**Unrestricted Client-Side API Key Management** Gemini API integration uses standard keys that lack granular service-bound credentials or rate limits.	Exposes the application to potential API quota exhaustion or security risks under higher loads.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance Uvicorn server running locally ([http://127.0.0.1:8000]	Deploy containerized FastAPI application with Docker on AWS Elastic Container Service (ECS) or Google Cloud Run behind an Application Load Balancer with auto-scaling.

		(http://127.0.0.1:8000)).	
2	Data Storage	No active persistent storage; responses are passed dynamically to index.html.	Integrate PostgreSQL for user profiles and learning histories, paired with Redis for caching generated summaries and quiz JSON responses.
3	Performance	Local PyTorch model CPU inference (LaMini-Flan-T5-783M) and direct synchronous API calls.	Offload local models to dedicated cloud GPU endpoints (e.g., Hugging Face Inference Endpoints or AWS SageMaker) and utilize async task queues (Celery) for heavy jobs.
4	Security	API keys managed directly via local environment setup (GEMINI_API_KEY).	Migrate to Gemini Authorization (auth) Keys bound to Google Cloud Service Accounts, backed by Google Cloud Secret Manager and CORS middleware configuration.

Future Roadmap:

Phase	**Planned Feature / Enhancement**	**Target Timeline**	**Expected Impact**
Phase 2	**Multimodal Input Support (PDF & Vision)** Allow users to upload PDFs or image snapshots of textbook pages/notes directly through the interface.	Q2 2026	Enables snapshot-based doubt solving, PDF summarization, and direct quiz generation from user-uploaded study materials.
Phase 3	**Voice-Based Interaction & Mobile Application** Add speech-to-text and text-to-speech capabilities alongside a cross-platform mobile app.	Q3 2026	Expands hands-free learning accessibility and provides anytime-anywhere study support with offline caching.
Phase 4	**LMS Integration & Teacher/Student Dashboard** Connect with platforms like Google Classroom/Moodle and introduce progress tracking analytics.	Q4 2026	Facilitates classroom collaboration, structured group study sessions, and visual learning progress metrics.